Forecast the chute. Inform the operator.

GRU-Based Chute-Level Forecasting & Carrier-Speed Advisory

DEEP LEARNING PROJECT



INPUTS (Electronically recorded mill process data)

Carrier drive records
RPM or Hz • Current • Voltage

Electronic chute level
Sensor readings • Quality flags

Shredder & context
Steam flow • Run/fault status

PROPOSED WORKFLOW

1	2	3	4	5	6
Mill records	Time split	History	GRU forecast	Advisory	Display

MODEL — 12 steps × 5 s → GRU (32 units) → Dense (16, ReLU) → Level (%) at +30 s

Initial configuration; timing refined to actual records.

ILLUSTRATIVE UI • RECORDED REPLAY

Current level
68%

Predicted level +30 s:
72%

Target band
60 – 75%

Forecast interpretation

WITHIN TARGET BAND

72% at +30 s

Hold current feed setting

Example carrier • demonstration settings

Current	Recommended	Change
40 Hz	40 Hz	0 Hz

Action: **HOLD**

Reason: Forecast within target band

Input quality: **Fresh**

TWO TEST MODES

Recorded replay
Held-out mill records • Forecast accuracy

Generated scenarios
Supplemental response & robustness tests

COMPARISON & EVALUATION

Compare:
Persistence • Linear • MLP • LSTM

Evaluate:
MAE/RMSE • Warning precision/ recall • Lead time • CPU latency

Operator review • Verified limits required for real settings • No drive commands

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Python • PyTorch • Time-series forecasting
Concept display and illustrative values